

Muchu Venkata Karthik

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Summary

Backend-focused Software Engineer with hands-on production experience engineering scalable microservices, REST APIs, and database architectures using TypeScript, Node.js, and PostgreSQL. Demonstrated background in building distributed schedulers, asynchronous task pipelines, and secure role-based access systems, with practical exposure to Redis caching and high-throughput data processing.

Skills

Languages JavaScript (ES6+), TypeScript, SQL, Java, Python
Backend & Systems Node.js, Express.js, Microservices, RESTful APIs, WebSockets, Background Workers
Databases & Caching PostgreSQL, MongoDB, Redis, Supabase
Frontend React.js, Next.js, Tailwind CSS, HTML5, CSS3
Tools & Architecture Git, GitHub Actions, Cloudflare (Workers, API Proxy), Postman, Winston

Experience

Full-Stack Web Development Intern

Feb 2026 – July 2026

Webileapps

Vijayawada, India

- Engineered backend microservices and full-stack modules using JavaScript, TypeScript, Node.js, and React across 3 internal enterprise fintech platforms handling multi-role access control.
- Designed and optimized relational schemas in PostgreSQL and document collections in MongoDB, ensuring transactional data integrity and reducing redundant queries.
- Integrated Redis caching and streamlined inter-service communication to reduce API latency by caching frequently accessed permission schemas and session records.
- Configured centralized audit logging with Winston and implemented JWT-based authentication with strict payload validation to safeguard financial data integrity.

Projects

API Coolie 🍷 | Node.js, JavaScript, MongoDB, Redis, Cloudflare API | Distributed System

Mar 2026 – July 2026

- Architected a distributed API scheduler and automated script execution engine across 40+ REST endpoints, executing scheduled HTTP webhooks and isolated scripts via custom cron timers.
- Engineered an asynchronous task-dispatch pipeline with dynamic payload parsing, automated retry strategies, idempotency safeguards, and real-time execution status polling.
- Integrated OpenRouter LLM API to provide automated failure diagnosis and root-cause analysis on unsuccessful job runs.
- Managed DNS routing, custom reverse proxies, and rate-limiting rules via the Cloudflare API for the primary domain.

TrackWicket 🍷 | Node.js, React.js, REST API, OneSignal | Full-Stack Platform

Jan 2025 – Feb 2025

- Constructed a real-time cricket data aggregation platform utilizing resilient background web scrapers and RESTful endpoints to process live match feeds, ICC standings, and player profiles.
- Optimized asynchronous state synchronization on the client and engineered structured API caching to eliminate unnecessary network round-trips during live matches.
- Implemented automated event-driven push notifications using OneSignal webhooks for boundary, wicket, and match milestone alerts.

Fourzdeals 🍷 | MERN Stack, Payment Gateway | Full-Stack Application

Mar 2023 – May 2023

- Developed an e-commerce platform featuring multi-tenant role-based authentication, structured product inventory schemas, and automated transactional invoice generation.
- Integrated payment gateway webhooks with cryptographic signature verification to ensure idempotent order fulfillment and secure checkout.

Additional Work: Box office analytics pipeline: reverse-engineered undocumented APIs, built Cloudflare Workers as edge proxy interceptors with custom session headers, and automated real-time seat tracking | Production Recipe Manager backend (TypeScript, MS SQL) | WebSocket communication service

Education

B.Tech in Computer Science and Engineering

2022 – 2026

Koneru Lakshmaiah Education Foundation

Guntur, Andhra Pradesh

- CGPA: 9.2 / 10.0
- Relevant Coursework: Database Management Systems, Data Structures & Algorithms, Operating Systems, Computer Networks

Certifications

- Wipro TalentNext Certified Java Developer
- Automation Anywhere Certified Essentials RPA Professional 🍷